



To perform the first successful blood transfusion, James Blundell took blood from the arm of his assistant and injected it into the patient.

THE BRITISH PHYSICIST

MICHAEL FARADAY (1791–1867), later famous for his work on electromagnetism, spent the initial years of his career concentrating on chemistry. Together with the utensil maker James Stodart, Faraday began to experiment with different **steel alloys** to incorporate rare metals such as platinum. These new alloys were too expensive to be

commercial, but they showed the value of a scientific approach.

The demands of industry accelerated progress in technology, and the increased status of engineers was reflected in the founding of the Institute of Civil Engineers in London. Steam locomotives were becoming more than curiosities, but they were still expensive to run and liable to explode. British engineer

Robert Stirling invented a **heat engine** that was intended as an alternative. His engine worked by continually **compressing and expanding air** or another gas in a closed space. The Stirling engine didn't catch on at the time although recently it has excited interest as a simple and low maintenance power source for everything from use in the Third World to space exploration.

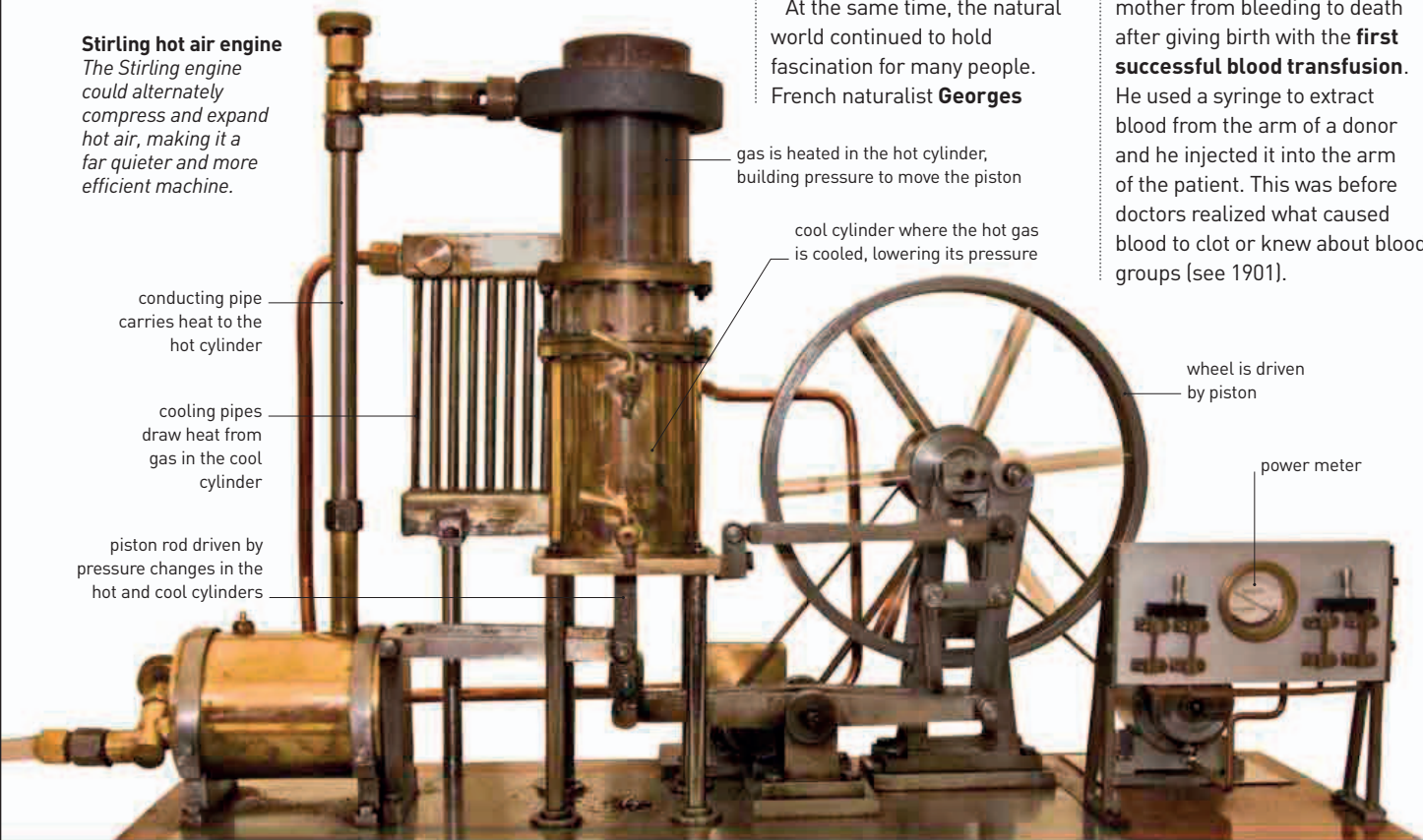
At the same time, the natural world continued to hold fascination for many people. French naturalist **Georges**

Cuvier studied fossils in the possession of the British clergyman William Buckland, found near Stonesfield in England a few years earlier. Cuvier confirmed that these fossils belonged to a gigantic extinct lizard.

This period also saw the growing professionalism of surgeons and doctors. London doctor **James Blundell** saved a mother from bleeding to death after giving birth with the **first successful blood transfusion**. He used a syringe to extract blood from the arm of a donor and he injected it into the arm of the patient. This was before doctors realized what caused blood to clot or knew about blood groups (see 1901).

Stirling hot air engine

The Stirling engine could alternately compress and expand hot air, making it a far quieter and more efficient machine.



gas is heated in the hot cylinder, building pressure to move the piston

cool cylinder where the hot gas is cooled, lowering its pressure

conducting pipe carries heat to the hot cylinder

cooling pipes draw heat from gas in the cool cylinder

piston rod driven by pressure changes in the hot and cool cylinders

wheel is driven by piston

power meter



William Parry's ships *HMS Hecla* and *HMS Griper*, trapped in the Arctic ice.



hole placed on patient's body

Stethoscope
Leaennec's stethoscope allowed doctors to hear murmurs inside a patient's chest.

earpiece

ANOTHER MEDICAL ADVANCE

was the **simple stethoscope**, invented by Parisian physician **René Laennec** (1781–1826) in 1816 to listen to the heartbeat and breathing patterns. It was first described in his 1819 book on diagnosis, *De l'Auscultation Mediate (On Mediate Auscultation)*. His stethoscope allowed physicians to **diagnose diseases sooner and far more accurately**.

Also in Paris, French physicists **Alexis Petit** (1791–1820) and **Pierre Dulong** (1785–1838) found a way to **verify the atomic weights of elements**. In 1803, British chemist **John Dalton** had put forward an atomic theory, stating that each element is made of atoms of a particular weight, but establishing the

British engineer
Robert Stirling invents
the **Stirling engine**

London doctor James
Blundell carries out the
first successful **blood
transfusion**

British scientist
Michael Faraday
experiments with
steel alloys

French chemist Louis
Jacques Thénard
discovers **hydrogen
peroxide**

French
physician René
Laennec describes
his invention of the
stethoscope in *De
l'Auscultation
Mediate*